

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Mutation

Replacement of nitrogenous base with another in one or both the strands

OR

Addition or deletion of a base pair



Permanent changes in the nucleotide sequence of DNA

Mutation

- ❖ One of the characteristics of genetic material
- ❖ Failure to store the genetic information.
- ❖ Mutation will affect on the gene expression and is propagated by replication
- ❖ Alteration of information within a single gene
- ❖ The term “point mutation is also used and is more specific
- ❖ As the change involves a single nucleotide change

Classification of Mutations:

a. Origin

1. Spontaneous mutation
2. Induced mutation

b. Cell type

1. Somatic mutation
2. Gametic mutation

c. Type of change

1. Substitutional mutation
2. Frameshift mutation

a. Origin

1. Spontaneous mutation

2. Induced mutation

- These are considered as arising in nature.
- No specific factors, other than natural elements, are associated with their occurrence. Arise by chance (random error in the storage of genetic information)
- Causes: exposure to background radiation (cosmic rays or radioactive minerals, chemical and thermal elements)
↓
Result in error during replication → Error in genetic code
- The enzyme of DNA replication (DNA polymerases) make error during replication
- Has been estimated to average about once in 100 millions (10^{-8}) cell divisions.
- Mutation rate is higher in Drosophila and humans (10^{-6} to 10^{-5}) per gamete formed.

2. Induced mutation



- arise as the result of the influence of any artificial factor.
- as (abnormal environmental factors as ionizing and non-ionized radiations or chemical mutagens)

b. Cell type

1. Somatic mutation

- Happen during replication of DNA in somatic cells.
- They are not transmitted to future generations
- Not impair the organism except if its cancer cell

✓ 2. Gametic mutation

- Have the potential of being expressed in all cells of an offspring
- Also may be transmitted to future generations, gradually increasing in frequency within the population
- Dominant autosomal mutations will be expressed phenotypically in the first generation
$$\begin{array}{ccc} \underline{aa} & \times & \underline{aa} \\ & & \rightarrow \underline{Aa} \times \underline{aa} \\ & & \underline{Aa} \end{array}$$
- Recessive autosomal mutations not expressed in F1
- Sex linked recessive mutation in heterogametic female → transmitted in hemizygous male offspring

x^ax^a

x^Ax^A

x^Ax^a

x^Ay

x^ay

homogametic

heterogametic

hemizygous

Mutations: Types

Base Substitution or Point Mutation

Replacement or change in a single base

Transition

Purine $\leftrightarrow$ Purine

Pyrimidine $\leftrightarrow$ Pyrimidine

T $\leftrightarrow$ C

A $\leftrightarrow$ G

Transition mutation

Frame Shift Mutation

T $\rightarrow$ A A $\rightarrow$ T
C $\rightarrow$ G G $\rightarrow$ C

Transversion mutation

Transversion

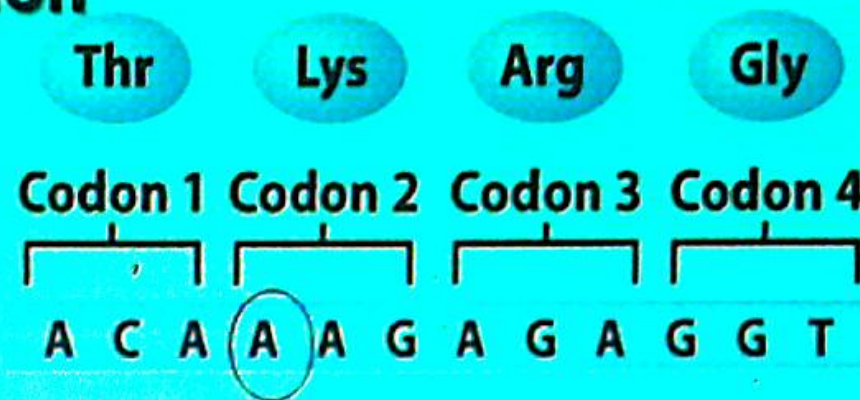
Purine $\leftrightarrow$ Pyrimidine

Pyrimidine $\leftrightarrow$ Purine



2. Frameshift mutation

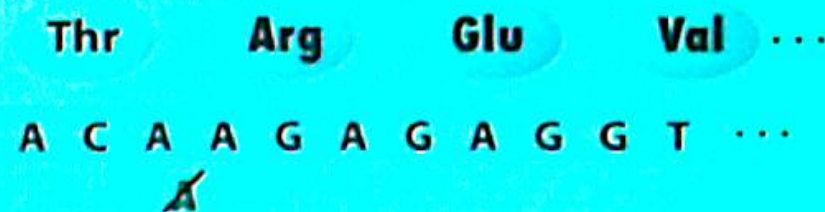
Wild-type gene



Gene with insertion



Gene with deletion



2. Frameshift mutation

- Caused by the insertion or deletion of a single nucleotide at any point along the gene



shift in the reading frame of **the code**



The **polypeptide chain** becomes completely **disrupted** after the location of change, thus creating a **nonsense mutation**

Frameshift Mutations:

- ❑ insertion or deletion of individual nucleotides and cause the rest of the message downstream of the mutation to be read out of phase, producing an incorrect protein from then on.

Types of point mutation

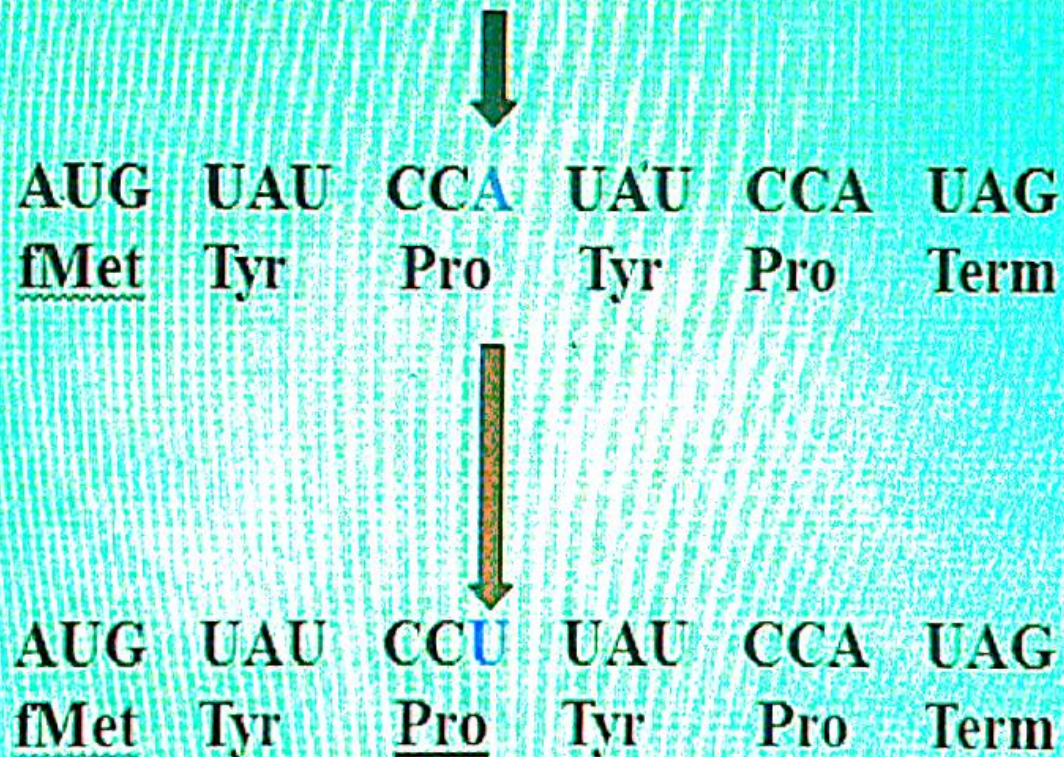
1- Silent Mutation

2- Missense Mutation

3- Nonsense Mutation

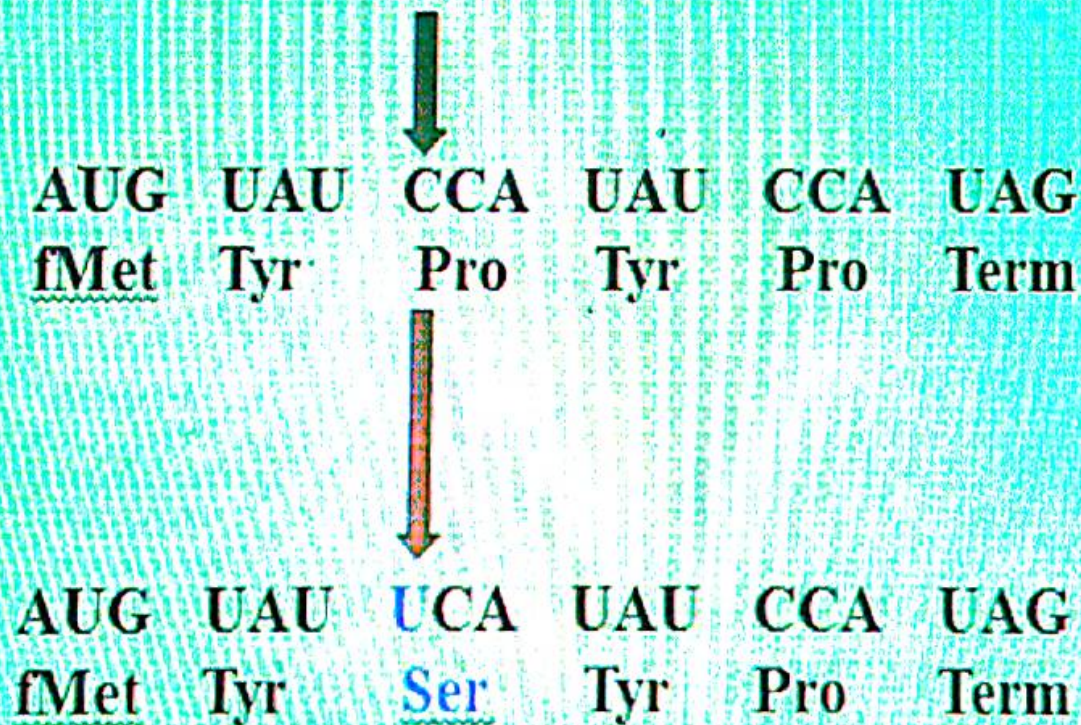
4- Fram shift Mutation

1- Silent Mutation



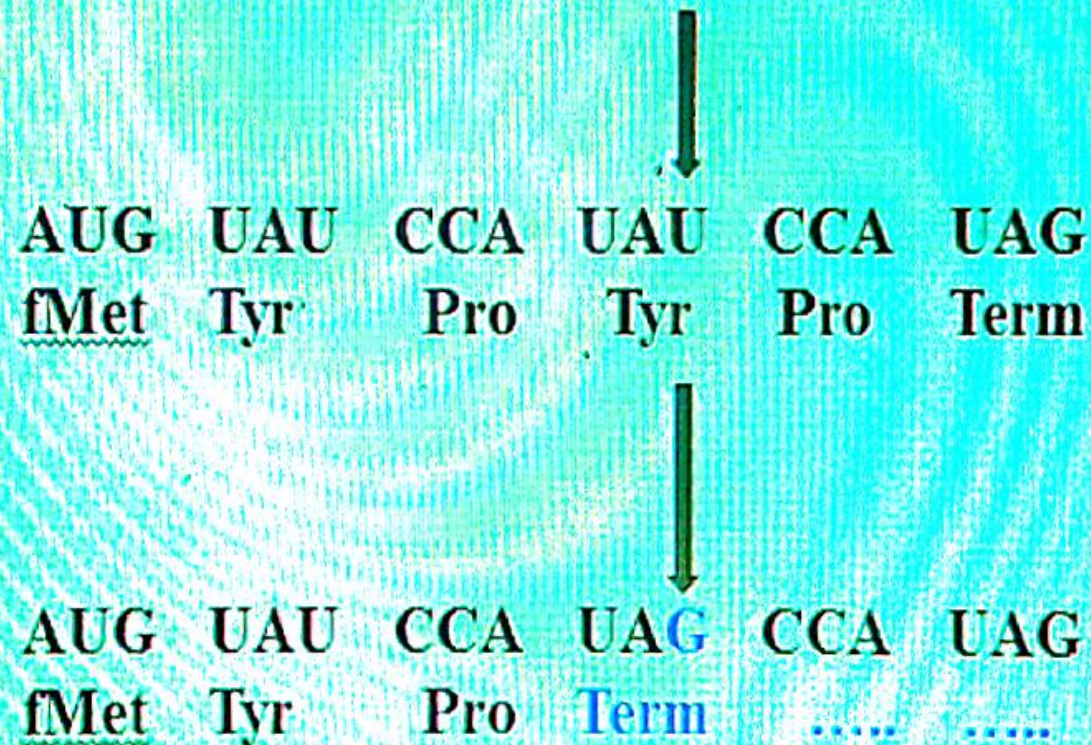
Silent Mutations change the nucleotide but not the amino acid sequence because they affect the third position of the codon

2- Missense Mutation



Missense Mutations change the meaning of a codon, changing one amino acid into another

3- Nonsense Mutation



Nonsense Mutations arise when a codon for an amino acid is mutated into a **termination codon** (UAG, UAA, or UGA), resulting in the production of a shorter protein.

4- Fram shift Mutation

Insertion

Deletion

ex.

AUG UAU CCA UAU CCA UAG
fMet Tyr Pro Tyr Pro Term

AUG UAU **GCC** AUA UCC AUA
fMet Tyr Ala Ile Ser Ile

Missense mutation

UAU → UCU
Tyr Ser

Nonsense mutation

UAU → UAA
Tyr Stop

Silent mutation

UAU → UAC
Tyr Tyr

Frameshift mutation

Original base sequence ATG AAA GAG UAU
Met Lys Glu Tyr

Base addition ATG AAG AGA GUA U
Met Lys Arg Val

Base deletion ATG AAG AGU AU
Met Lys Ser

Codon	Amino acid	Type of mutation
GAA → Glu		Silent mutation
GAG → Glu		
GAA → Glu		Nonsense mutation
UAA → Stop		
GAA → Glu		Missense mutation
GAC → Asp		

❖ Nonsense and fram shift Mutation → **Never go**
unnoticed (give shorter or abnormal protein
which are in active)

❖ Silent Mutation → unnoticed

❖ Missenes Mutation → give some degree of biological
activity

Mutagens

Mutagens → are agents which cause a **direct or indirect change** in the base sequence in a chromosomal DNA

✓ Ways of change occurrence:-

- 1- **Incorrect** insertion of a **base** during replication which is not removed
- 2- Insertion of a rare **tautomeric form** of a base
- 3- **Chemical alteration** of a **previously inserted base** to one which has a different base-pairing specificity
- 4- **Deletion** of one or more **bases** during replication
- 5- Insertion of one or more extra **bases** during replication

Mutagens

Mutagens includes a variety of environmental factors which divided into:

1. Chemical mutagens

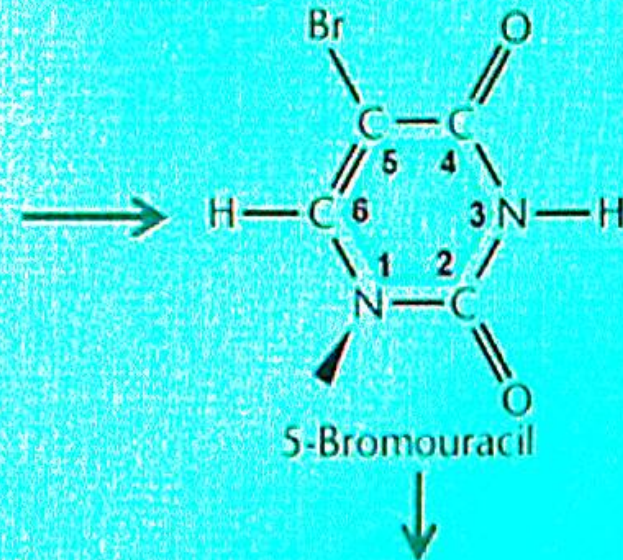
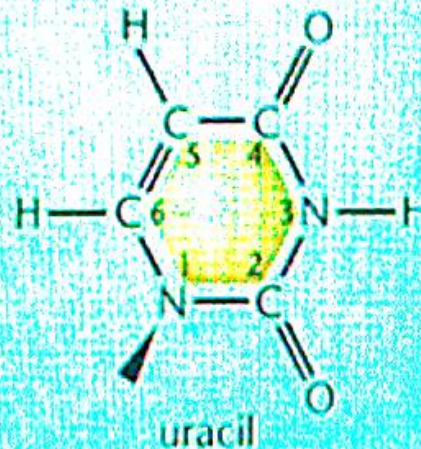
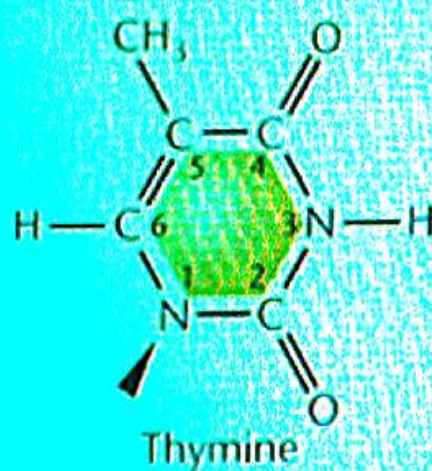
- Base Analogues
- Nitrous Acid
- Hydroxylamine
- Alkylating Agents
- Acridine Dyes

2. Radiation mutagens

1. Non-ionizing radiation:
 - Heat
 - Ultraviolet Radiation (UV)
2. Ionizing Radiation
 - X rays
 - Cosmic rays

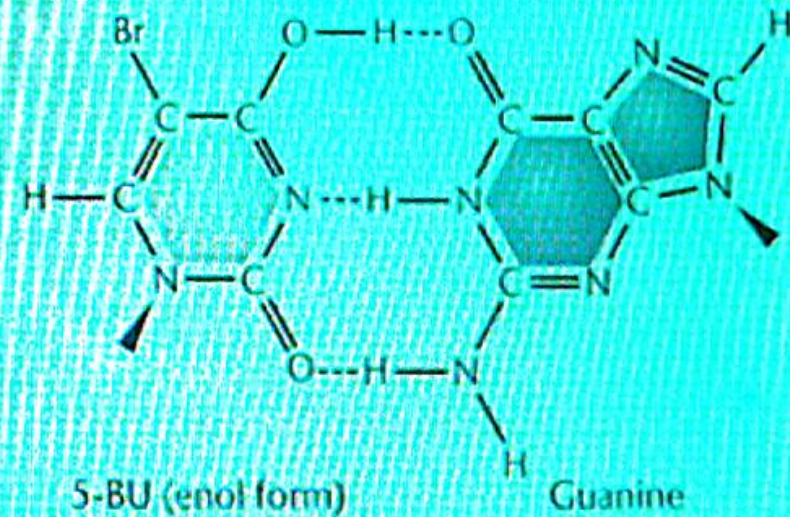
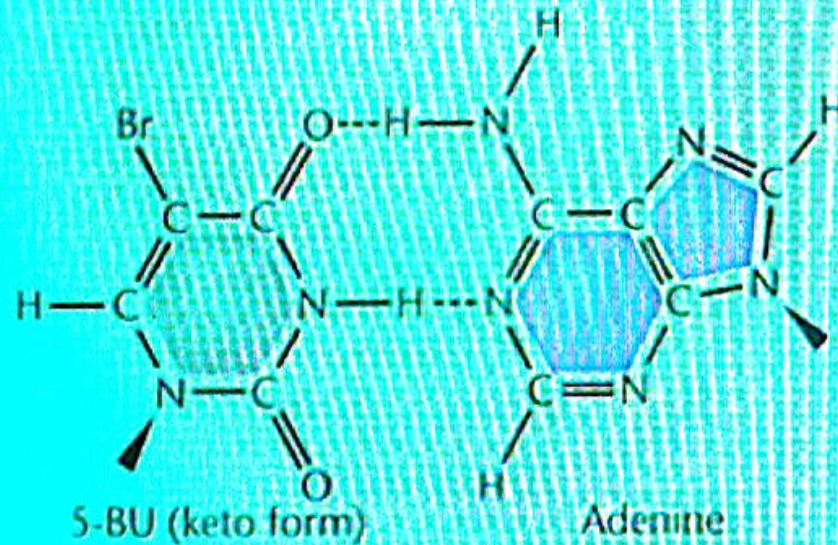
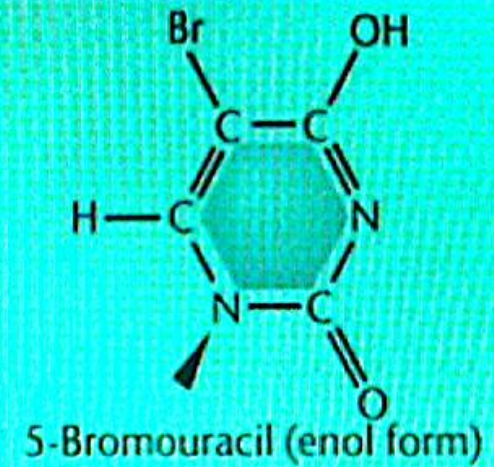
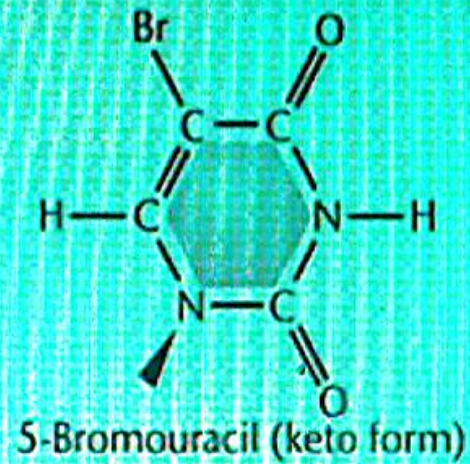
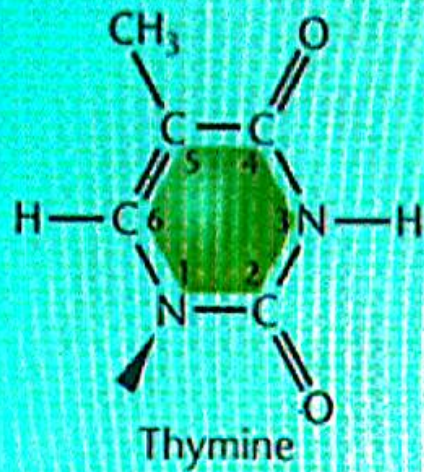
a. Base Analogues → are substitutes of purines or pyrimidines during DNA synthesis

- They include the **halogenated derivatives** of uracil in the number-5 position of the pyrimidine ring: as in 5-bromouracil, 5-chlorouracil and 5-iodouracil



5-BU is incorporated into DNA in place of T
So, transition from A-T to G-C occur

← increases the probability of
a tautomeric shift



1. Chemical mutagens

b. Nitrous Acid → Oxidation, of deaminating purines and pyrimidines
(cytosine and **adenine**)

This cause → conversion of the amino group (NH_2) of
cytosine and adenine to a keto group (O=)

Thus, cytosine $\xrightarrow[\text{acid}]{\text{nitrous}}$ uracil



Pairs with **adenine** instead of guanine



G-C to **A-T** transition results

adenine $\xrightarrow{\text{deaminated}}$ hypoxanthine



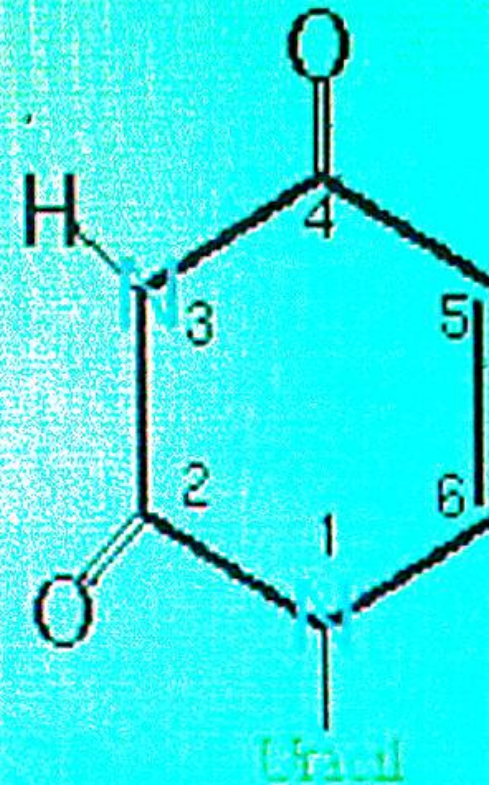
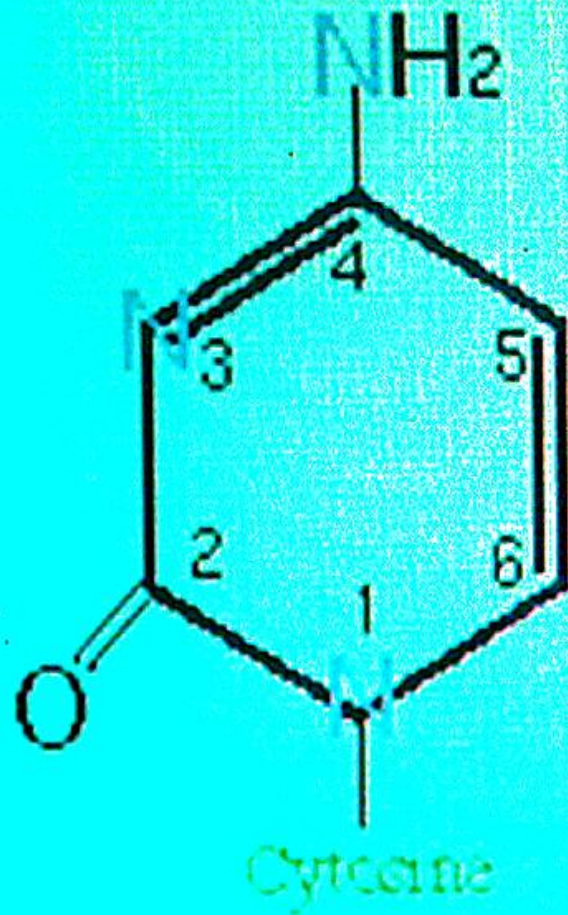
Pairs with cytosine instead of
its normal pairing with **thymine**



A-T to G-C transition results

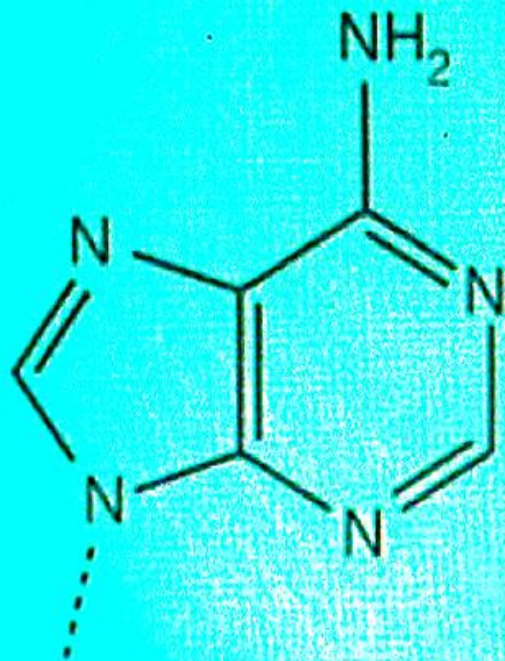
1. Chemical mutagens

b. Nitrous Acid $\longrightarrow$ Oxidative deamination

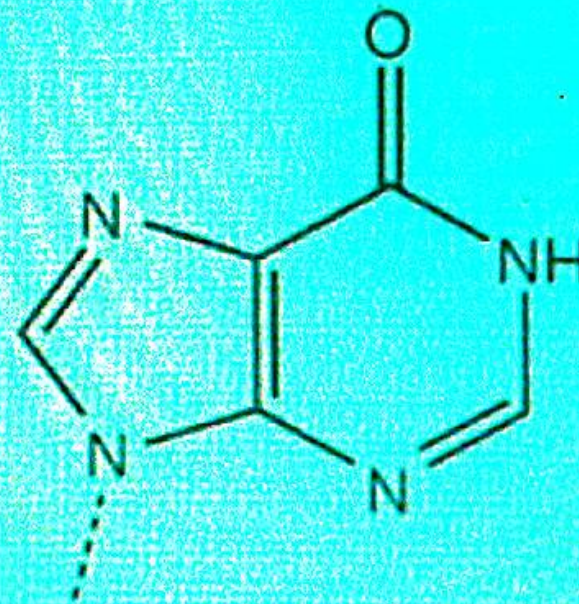


1. Chemical mutagens

b. Nitrous Acid $\longrightarrow$ Oxidative deamination



Adenine



Hypoxanthine

Oxidative deamination

1. Chemical mutagens

c. Hydroxylamine (NH_2OH)

Reacts specifically with cytosine by adding a hydroxyl group (OH) to its amino group

→ hydroxylaminocytosine



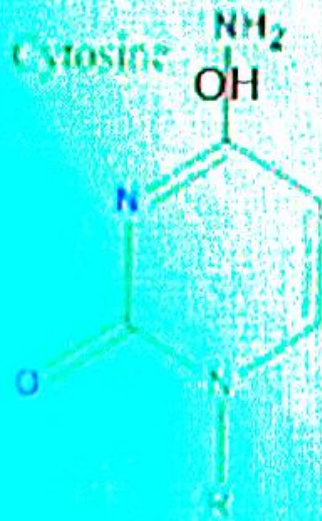
may now undergo a tautomeric shift



allowing it to pair with adenine instead of guanine



G-C pair is converted to an A-T pair

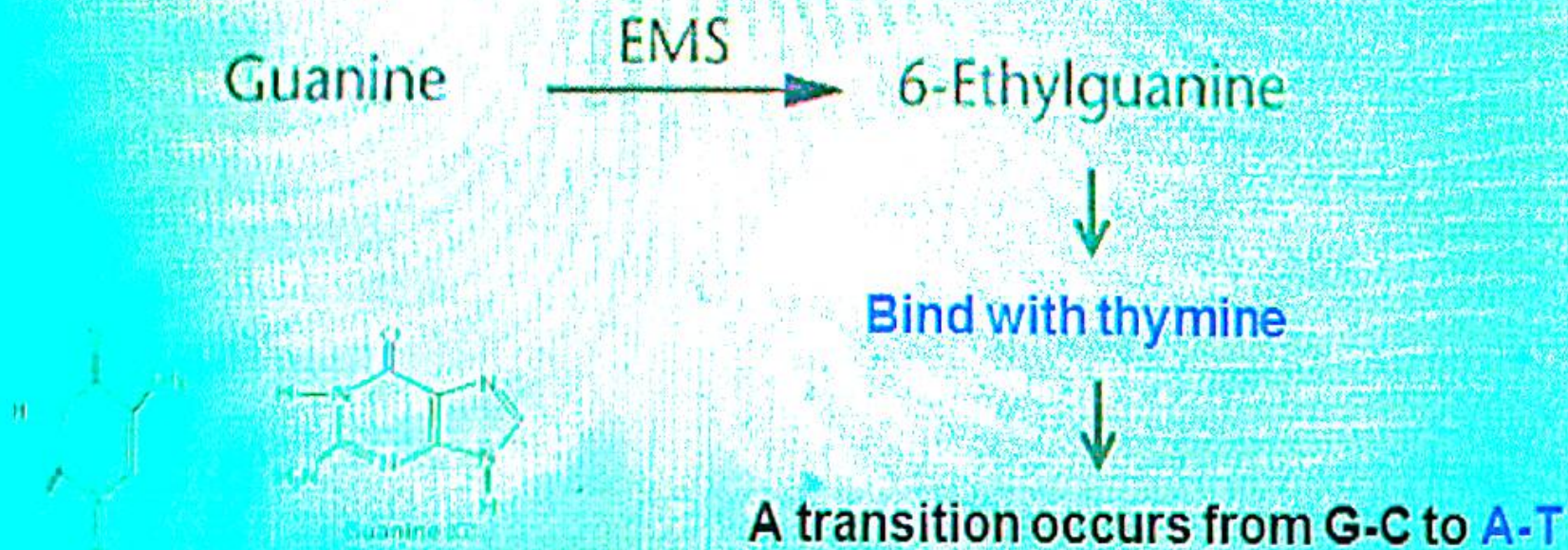


1. Chemical mutagens

d. Alkylating Agents → These include mustard gases

- Cause alkylation by donating an alkyl group such as CH₃- or CH₃-CH₂- to **amino** or **keto** groups in nucleotides

EX. Ethylmethane sulphonate (EMS) alkylates the **keto** group at the number 6 position of **guanine** and number 4 position of **thymine**



1. Chemical mutagens

e. Acridine Dyes (proflavine and acridine orange)

- Intercalate or wedge between purines and pyrimidines of intact DNA

➤ If intercalation occurs before replication



then the insertion of an extra base during replication leads to an insertion mutation



➤ if the intercalation occurs during replication



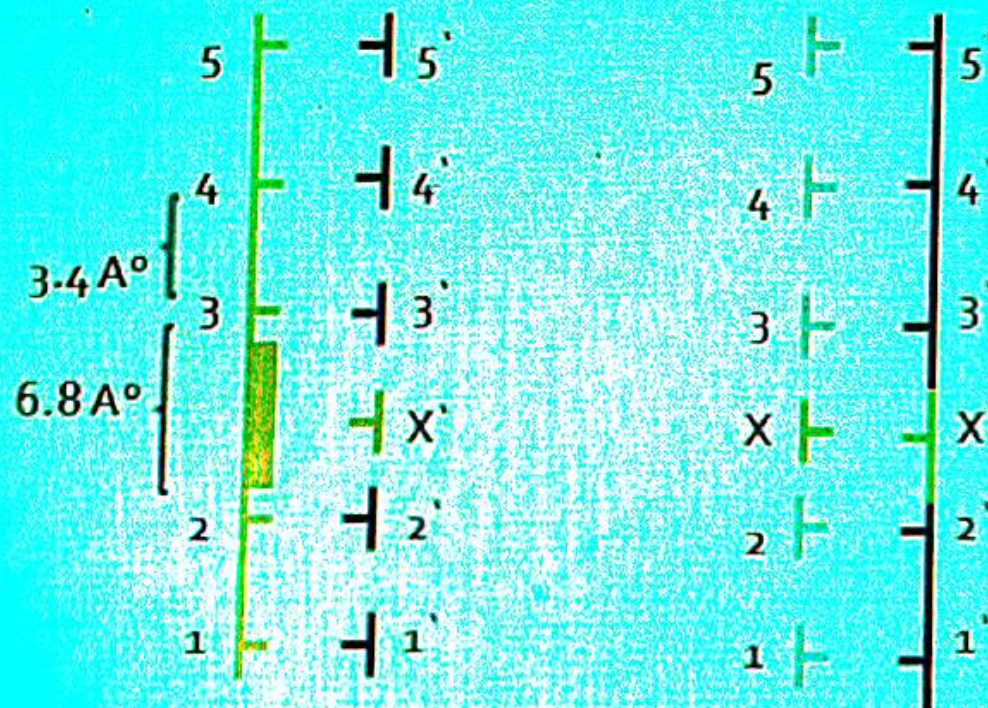
A base will be missing leading to deletion mutation



Both are considered Frameshift mutations

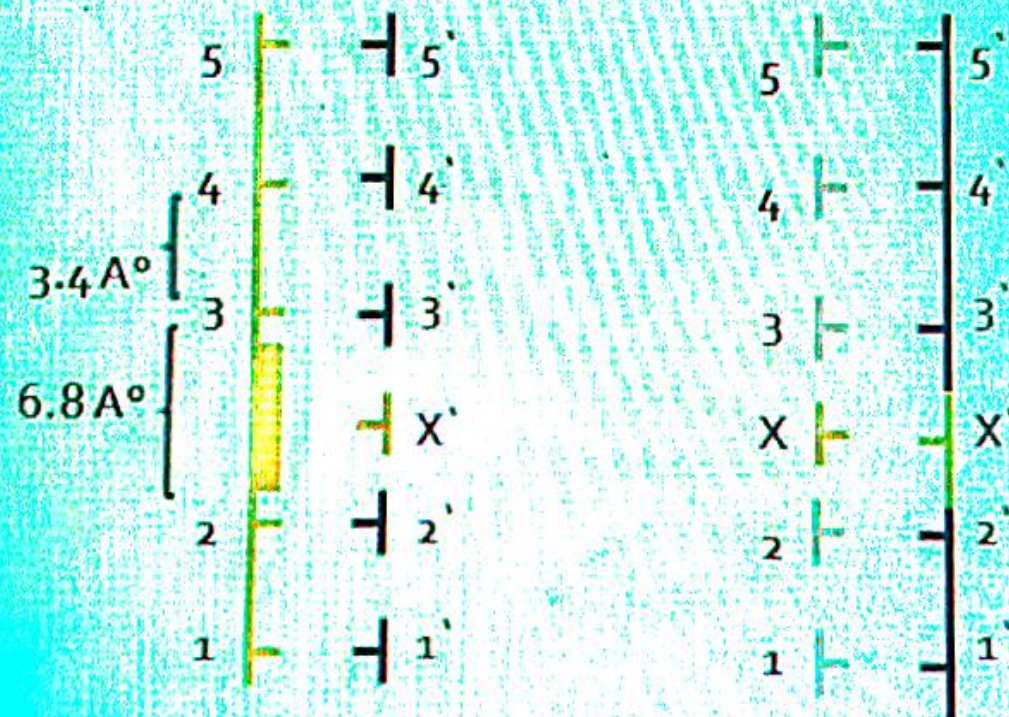
a. Insertion (Acridine intercalate before replication)

1st replication round 2nd replication round



a. Insertion (Acridine intercalate before replication)

1st replication round 2nd replication round



deletion (Acridine intercalation) **Thr** **Lys** **Arg** **Gly**

Wild-type gene

Codon 1	Codon 2	Codon 3	Codon 4
A C A	A A G	A G A	G G T

Gene with insertion

Thr	Glu	Glu	Arg ...
A C A	G A A	G A G	A G G T ...

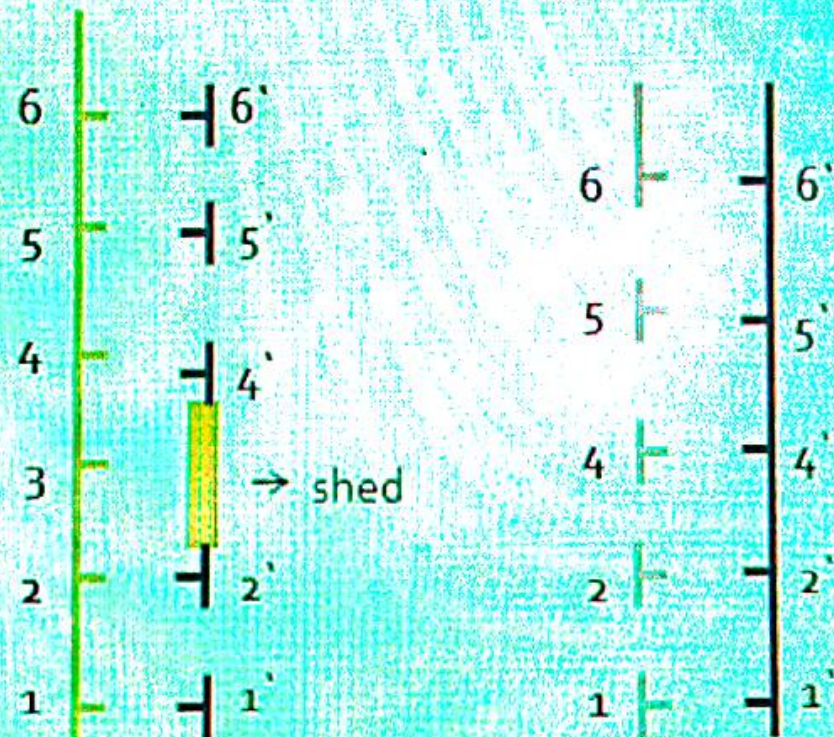
Gene with deletion

Thr	Arg	Glu	Val ...
A C A	A G A	G A G	G G T ...

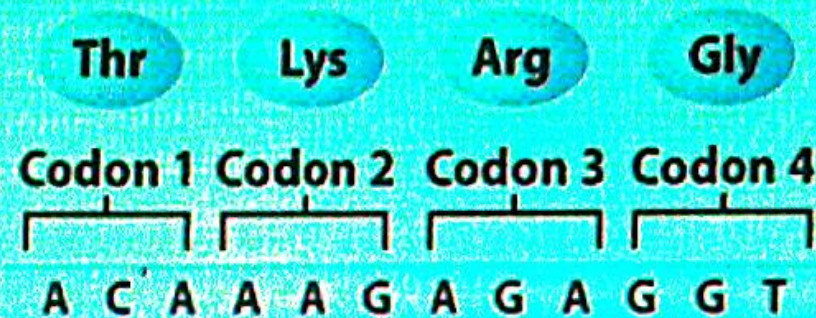
Both are considered Frameshift mutations

b. deletion (Acridine intercalate **during** replication)

1st replication round 2nd replication round



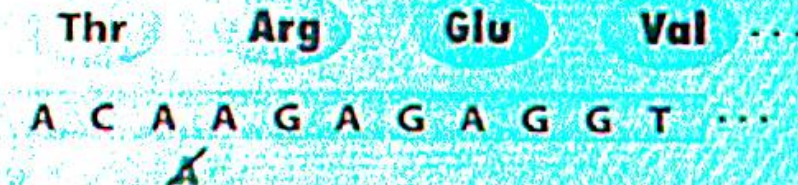
Wild-type gene



Gene with insertion



Gene with deletion



Both are considered Frameshift mutations

2. Radiation mutagens

- There are different types of radiation depending on the wavelength of the rays emitted
- Within the electromagnetic spectrum, the inherent energy varies inversely with wavelength

a. Non-ionizing radiation

- Longer wavelength
- Not strong enough to penetrate deeply into tissue
- It include heat and ultraviolet which affect mainly on surface tissue U.V rays Heat

b. Ionizing Radiation

- Short wavelength
- Strong and penetrate deeply into tissue and causing ionization of the molecules encountered a long their path
- Include x rays, gamma rays and cosmic rays

2. Radiation mutagens

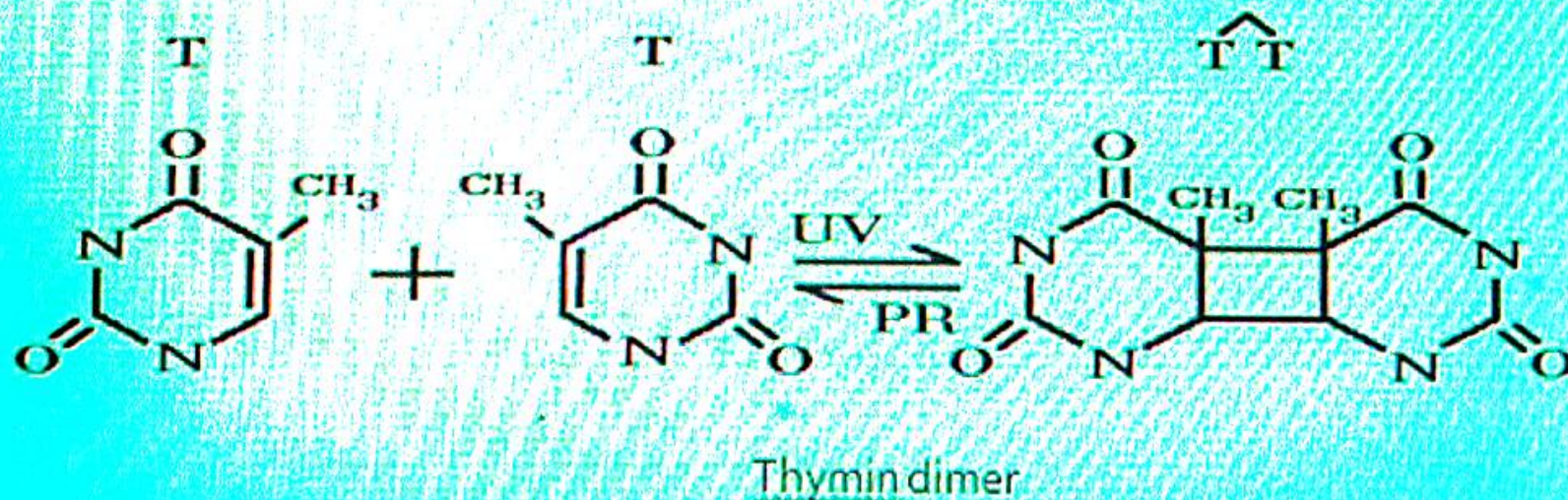
a. Non-ionizing radiation

1. Heat

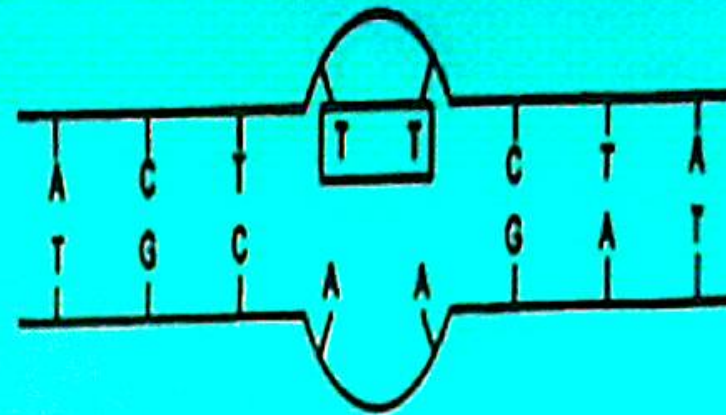
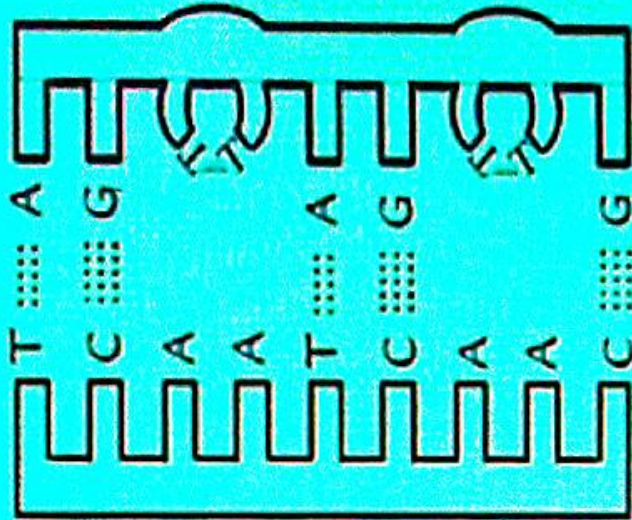
- Causes **deamination of cytosine** to form **uracil**, as does *nitrous acid*, thereby bringing about **transitions** from G-C to A-T
- also causes **traversions** from G-C to C-G by some unknown mechanism
- It has been estimated that **over 100 heat-induced mutations** occur in a typical human cell each day, **but** the vast majority of these are later repaired by one of the DNA repair mechanisms

2. Ultraviolet (UV)

- The major effect of UV light is on pyrimidines, leading to the formation of pyrimidine dimers
- UV radiation causes the formation of chemical bonds between two adjacent pyrimidine molecules, particularly between two adjacent thymine molecules, leading to thymine dimers

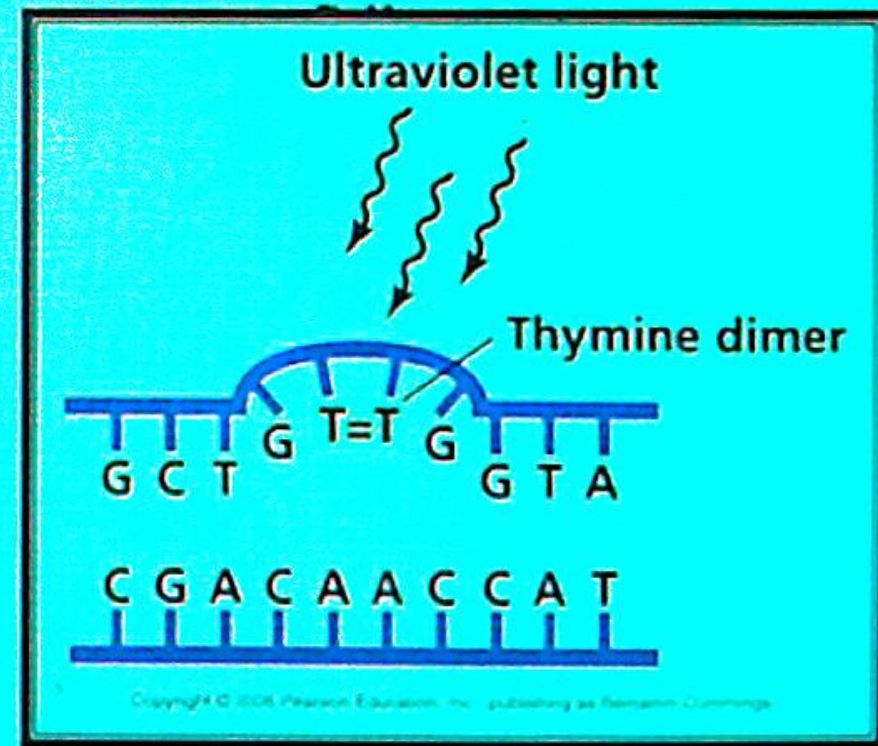


Thymine Dimers



Thymine dimer in one strand

Dimer distorted the DNA conformation and inhibit normal replication



- Dimerized bases → Cannot form hydrogen bonds with the complementary purines on the opposite DNA strand

- Dimerized → Inhibits normal replication and transcription



This alteration prevents the transmission of the altered genome from parent to offspring

The inhibitory effect of UV seems to be responsible for the killing effects of this radiation on microorganisms

2. Radiation mutagens

b. Ionizing Radiation

The effect of high energy rays on cells are believed to result from the rays colliding with atoms and molecules of the cells at high speed

Thus **stable molecules and atoms** are transformed to **free radicals** and electrically charged reactive ions



directly or indirectly affect the genetic material



altering purines and pyrimidines in DNA



resulting in **point mutations**

also capable of **breaking chromosomes**



a variety of aberrations

➤ **Experiments on Drosophila (effect of X-rays)**



a linear relationship exists between **X-ray dose level** and the **frequency of mutation**

- The effect of a **single large dose** (acute radiation) of ionizing radiation is more mutagenic than that of the **same dose divided into smaller exposure** (chronic radiation) ↘

(cosmic rays)

acute radiation → are mostly artificial such as **X-rays** or **nuclear explosion**
↘ **Development of cancer especially leukemia**

Mutation and cancer

Cancer → Disease associated with change in **genetic material** that lead to the inability of the cell to control its reproduction → lead to malignant tumours

Genetic material → main target in case of mutagenicity and carcinogenicity

- Cell division appears to be under the control of genetic material, and if a mutation affects the gene which controls mitosis, it might loss its function.
- Thus, the cell loses the control over cell division, and goes on
- divding indefinitely, i.e. it becomes a cancer cell.

Carcinogens

Certain factors can change normal cell to malignant cells

Carcinogens

1. Chemical

Have direct effect on DNA as:

- Asbestos
- Tar product
- Mustard gas
- Polyvinylchloride

2. Physical

Mutagenic radiation:

- UV
- X-ray
- Gamma ray

3. Biological

- Oncogenic Viruses
- Oncogenes

Carcinogens

1. Chemical Carcinogens

- Chemical substance to be carcinogenic, it has to be mutagenic, i.e. its effect on DNA is direct.
- Chemical carcinogens include asbestos, tar products, mustard gas, and polyvinylchloride

2. Physical carcinogens

- UV cause skin cancer in humans affected with xeroderma pigmentosum.
- X-rays leukemia (blood cancer),
- Larger percentage of ex-technicians working with X-rays die of leukemia more than unexposed people.
- Gamma rays produced by atomic fission are cancer producing.

3. Biological Carcinogens

1. Oncogenic Viruses

It's a type of RNA virus which transformed the normal cell to malignant cell

Ex.

In chicken → Rous sarcoma virus

In mice → - Gross leukemia virus
- Bittner mammary tumour

✓ Gene in viral genome



oncogene

2. Oncogenes

Discovered in DNA from cells of human cancer of urinary bladder which is similar in its nucleotide sequence to the oncogene in oncogenic viruses

Oncogene present in normal human cell transform cells to malignant cell and called **Proto-oncogene**



it activated to oncogene under the effect of carcinogenes (radition, chemical, viruses)



Thank You!

c. Type of change

1. Substitutional Mutation

- ❖ Substitution of a single different nucleotide pair

Point mutation

TTT

AAA

Change function of
protein (missense)

Give different a.a

Change code

Substitution

Transition

- ✓ if a pyrimidine replaces a pyrimidine
or purine replaces a purine

Transversion

- ✓ if a purine and a pyrimidine
are interchanged

Regulation of Protein Synthesis

Dr shymaa khattab

Regulation of Protein Synthesis

Genetic Studies → Not all genes present in a cell are active in directing specific process of protein synthesis

The genes can be switched on or off depending upon:

1. Position of a cell in the body
2. Stage of body development
3. The external environment

Genes in the body divided into two types:

Constitutive genes

- These genes are expressed *continuously* and their products are found in the cell in almost the same amount at all the time.

- Controls governing the expression of it are invariant

ex. Enzyme of glycolysis

Regulated *genes*

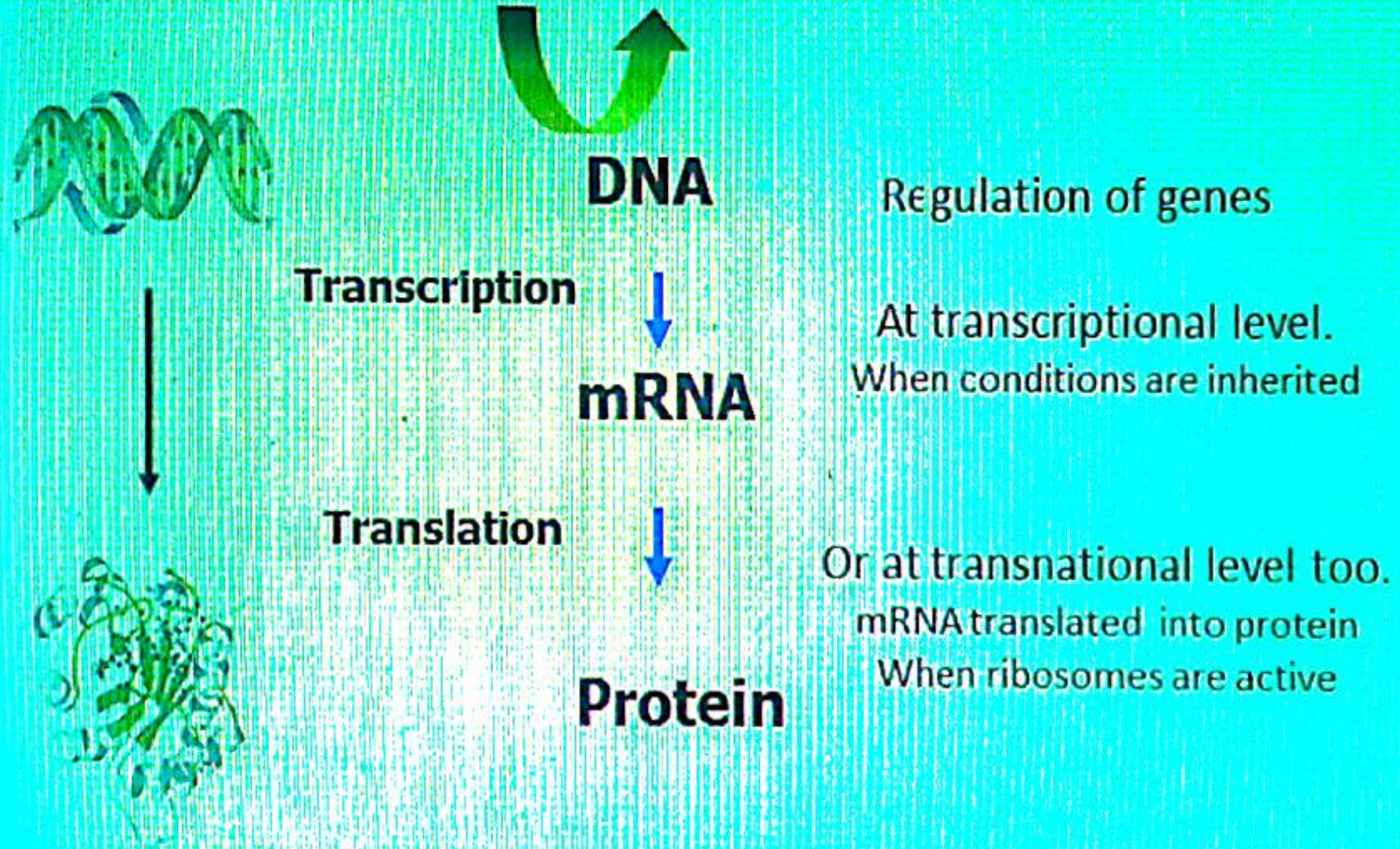
- Its expression are modified by the presence of absence of chemical stimuli in the genes environment

- On call, are expressed according to body needed so amount of its product differ from time to time

ex. Enzyme needed in the pathway for synthesis of a.a

Enzymes used in metabolism of Arabinose and galactose

Genetic expression Gene: DNA encoding proteins



***Thank you
for your attention***